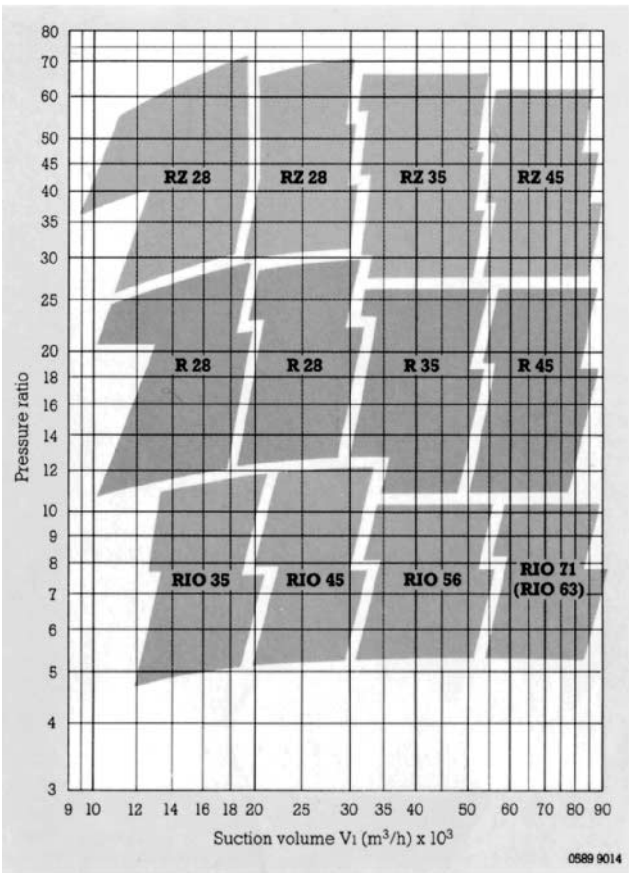
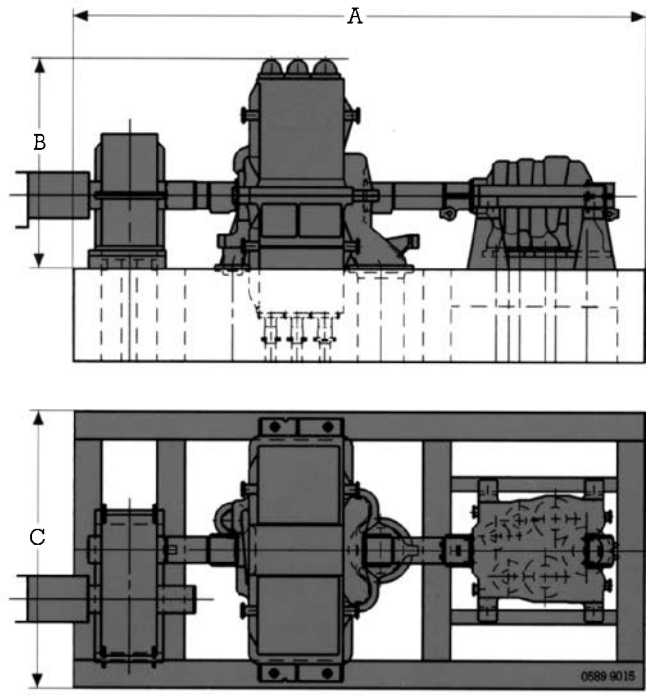


230



Dimensions for RIO + RZ compressor group

	A	B	C
RIO 35 + RZ 28	6200	2300	3000
RIO 45 + RZ 28	7000	2500	3200
RIO 56 + RZ 35	7200	2600	3200
RIO 63 + RZ 35	7300	2700	3300

231

Technical data (FIG. C-230) and dimensions (FIG. C-231) (RIO designation) of five frame sizes covering flow range from 12,000 to 90,000 m^3/h. (Source: Sulzer-Burckhardt.)

speed. The sense of rotation of the rotor is clockwise, seen from the suction side. See also Figs. C-243 and C-244.

Coolers. The three pairs of intercooler tube bundles are mounted in a vertical position on each side of the centrifugal casing and bolted to the lower water box. They can freely expand upward. The upper water box is fixed to the upper tube plate and guided in the top water chamber cover (Fig. C-245). Round finned tubes ensure excellent heat transfer on the air side. The gap between tube bundle and casing at the exit of the cooler is sealed with a rubber membrane to avoid bypassing of uncooled air. The cooling-water connection and condensate drains are located at the bottom.

Integrally geared compressors*

Multistage integrally geared compressors. The design principle of integrally geared centrifugal compressors was established, implemented, and patented by Demag

* Source: Demag Delaval, USA.

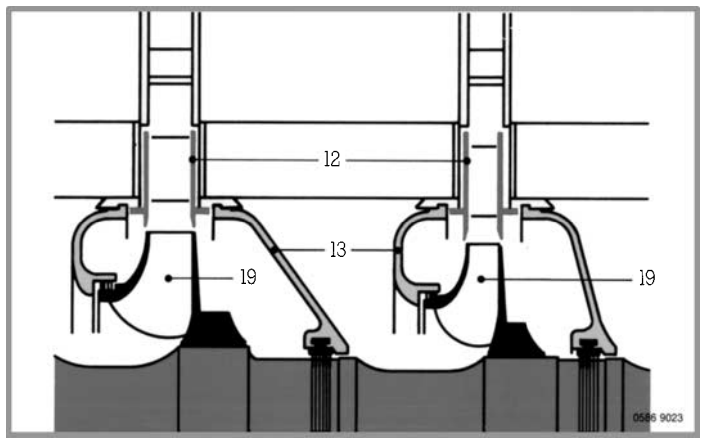


FIG. C-232 Centrifugal casing internals. (Source: Sulzer-Burckhardt.)

Delaval more than 50 years ago. Since then, this concept has been continuously further developed, enabling it to be successfully employed in demanding applications in process engineering. Figures C-246 through C-250 illustrate some typical applications. Figure C-251 illustrates the internal components of a multistage integrally geared compressor.

This information source's model VK/HVK integrally geared centrifugal compressors are characterized by a combination of several compression stages with one gear assembly, forming a compact unit for the multistage compression of a wide range of gaseous media.

Intergrally geared compressors offer the following advantages:

- Low power consumption due to concept, modern impellers, tailored aerodynamics, and optimized auxiliaries
- Wide control range and good part-load efficiency due to adjustable guide vane units in the first or on all compression stages
- Multiservice capability and many application options due to a system of standard components
- Package design available for almost all VK/HVK compressor sizes. A package includes compressor, process gas coolers, lube oil console, process piping, and all tubing and wiring, thus significantly reducing on-site installation time

Technical performance data (see Fig. C-252)

- Capacities from 1000 to 480,000 m^3/h (600 to 280,000 cfm)
- Discharge pressure up to 100 bar (1450 psi)
- Gearbox ratings up to 45,000 kW (60,000 hp)

Fields of application include

- Air separation
- Synthetic fibers
- Ammonia
- Ethylene oxide
- Hydrocarbons and more

Single-stage integrally geared or direct driven centrifugal compressors. Series SEZ/SEE process compressors represent the single-stage concept derived from multistage integrally geared VK/HVK machines.

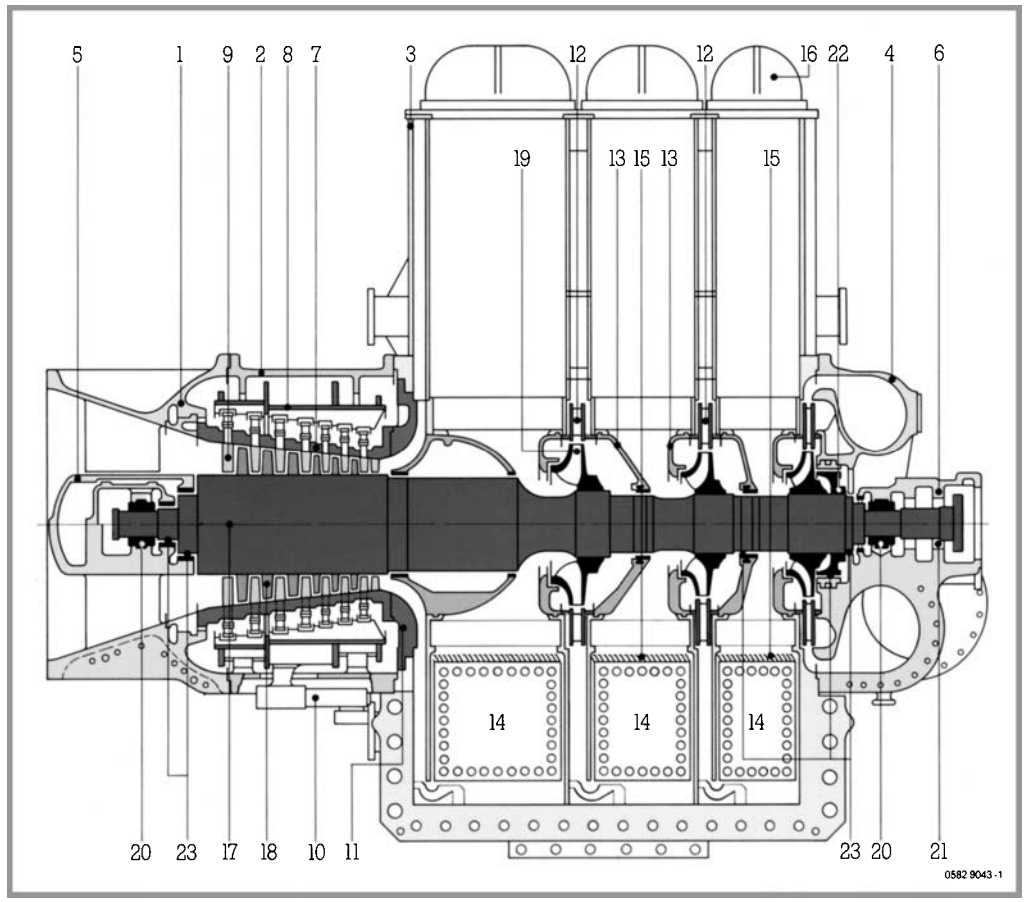


FIG. C-233 Section through an ARI series isotherm compressor (above, vertical section; below, horizontal section). 1, inlet casing of axial section; 2, center part of axial section; 3, casing of radial section; 4, discharge volute; 5, suction-side double-walled bearing housing; 6, bearing housing, discharge side; 7, blade carrier; 8, blade adjusting cylinder; 9, adjustable stator blades; 10, servomotor; 11, short-diffuser wall; 12, bladed diffusers; 13, partition walls; 14, cooler bundles; 15, water separator; 16, water chamber covers; 17, shaft; 18, rotor blades; 19, impellers; 20, journal bearings; 21, position of thrust bearing (if fitted); 22, balance piston; 23, shaft seal. (Source: Sulzer-Burckhardt.)

Components such as impeller volute casings, guide vane units, bearings, and seals are taken from standard components of that series.

The lube oil console is incorporated into the compressor base frame. This way compressor, driver, and auxiliaries form one single lift unit that is easy to install and easy to maintain.

While the SEZ series incorporates a stepup gear, the SEE has been designed for direct drive concepts.

Technical performance data

- Volume flow rates up to 48,000 m^3/h (280,000 cfm)
- Compression ratios up to 3 depending on the gas analysis
- Discharge pressure up to 80 bar (1200 psi)
- Discharge temperature up to 3000°C (572°F)

Fields of application include

- Petrochemical
- Refineries

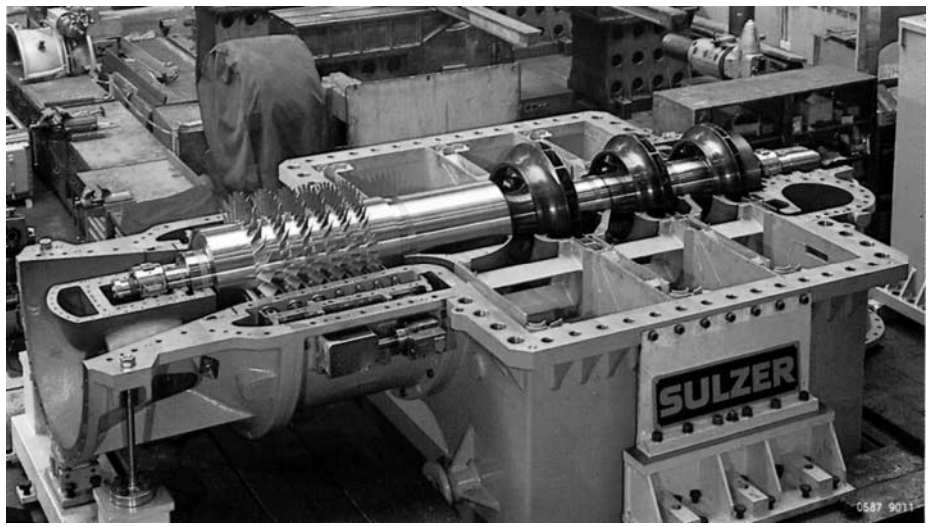


FIG. C-234 Functional and compact design with access to all vital components (ARI series). (Source: Sulzer-Burckhardt.)

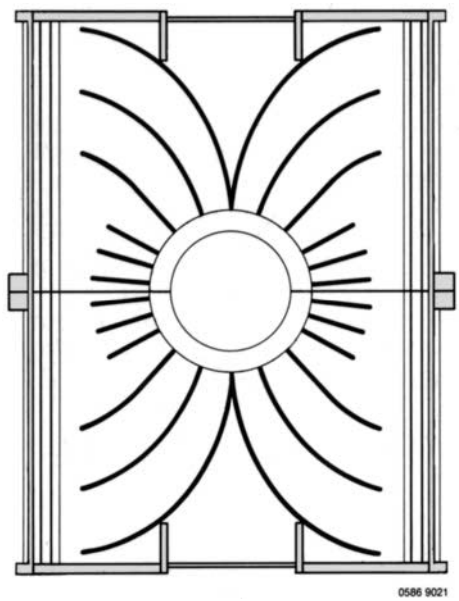


FIG. C-235 Flow passage between axial outlet and entry into the adjacent pair of intercoolers. (Source: Sulzer-Burckhardt.)

- Ammonia, fertilizers
- Metal production
- Air separation and more

See Figs. C-253 through C-259. Figure C-258 illustrates the internal components of a single-stage integrally geared compressor.

Integrally geared process gas radial turbines

An application for direct power recovery (see Figs. C-260 through C-262). The PRT series of integrally geared radial turbines has been developed on the basis of the integrally

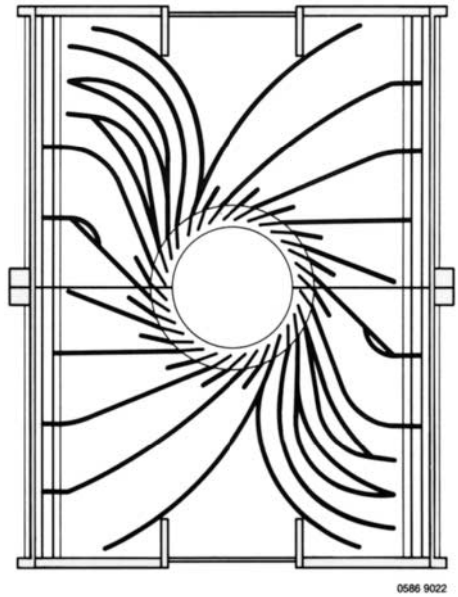


FIG. C-236 Flow passage from impeller outlet to the adjacent pair of intercoolers. (Source: Sulzer-Burckhardt.)

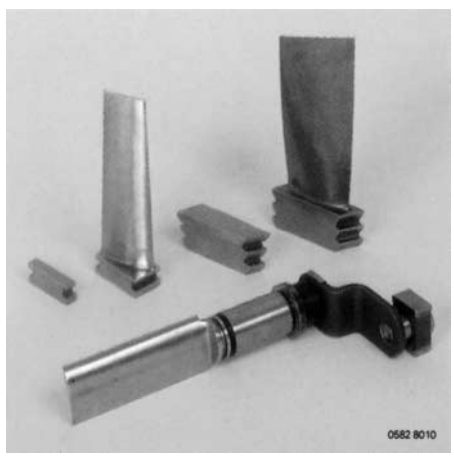


FIG. C-237 Adjustable stator blade, rotor blade, and fixed stator blade with intermediate piece. (Source: Sulzer-Burckhardt.)

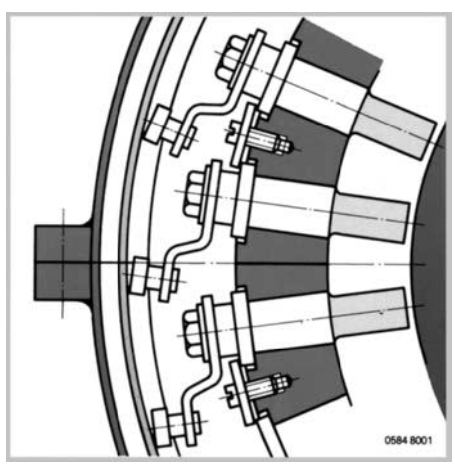


FIG. C-238 Fixation of adjustable stator blades. (Source: Sulzer-Burckhardt.)

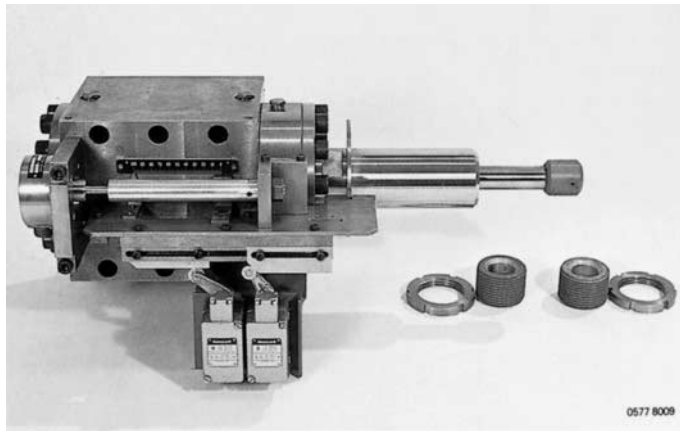


FIG. C-239 Hydraulic servomotor. (Source: Sulzer-Burckhardt.)

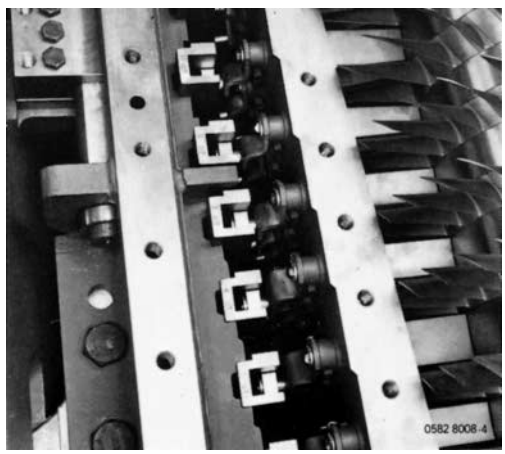


FIG. C-240 Stator blade adjusting mechanism. (Source: Sulzer-Burckhardt.)

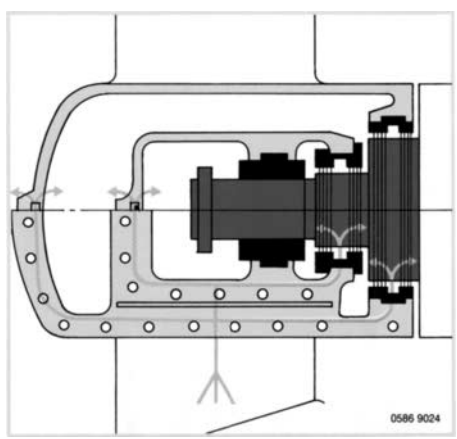


FIG. C-241 Double compressed-air sealing system on the suction side bearing housing. (Source: Sulzer-Burckhardt.)

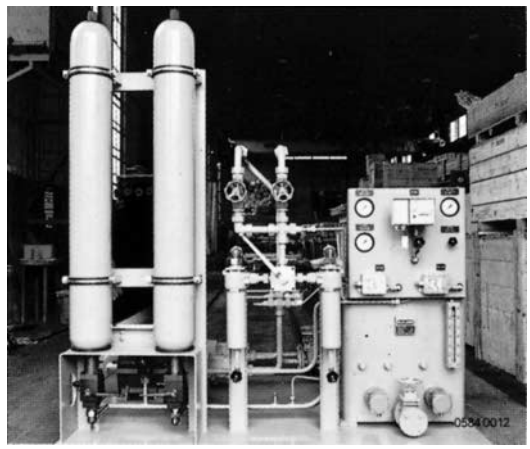


FIG. C-242 Power oil unit. (Source: Sulzer-Burckhardt.)

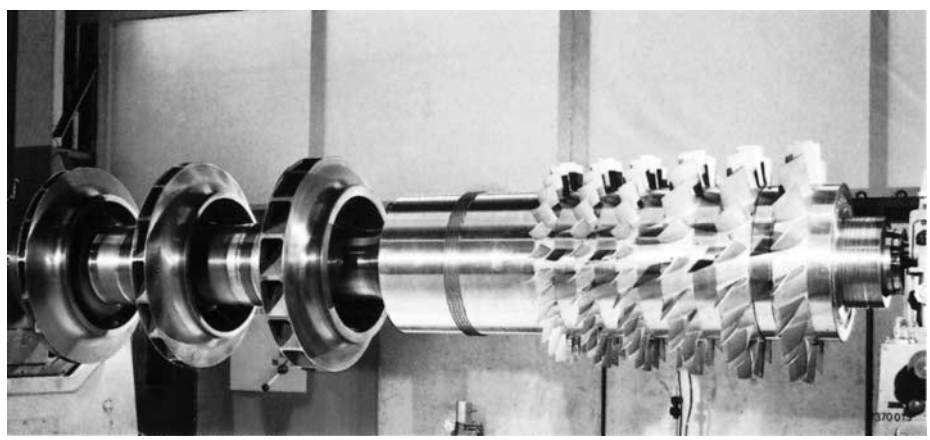


FIG. C-243 The single monobloc rotor running in only two journal bearings maintains high rotor stability and low vibration level. (Source: Sulzer-Burckhardt.)

geared centrifugal compressor concept using many of its components. Medium- to high-temperature process gas can be directly put to use driving a compressor or electric generator without the need for extra heat exchange or steam generation equipment.

Depending on the process conditions the PRT stages can either be mounted to the gearbox of a multiservice VK/HVK machine or designed as a separate unit.

For part-load operation the PRT can be fitted out with adjustable inlet nozzles.

Technical performance data (see Fig. C-263)

- Gas inlet temperatures up to 550°C (1020°F)
- Discharge volumes up to 350,000 m^3/h (200,000 cfm)
- Power recovery up to 35,000 kW (48,000 hp)

Fields of application include

- Power recovery
- Nitric acid and more

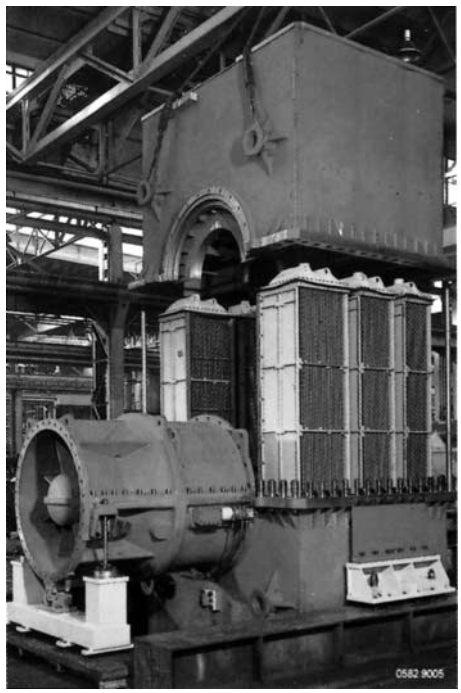


FIG. C-244 Inspection of the internal parts: Lift the centrifugal casing top half without disturbing coolers and adjacent casing parts. (Source: Sulzer-Burckhardt.)

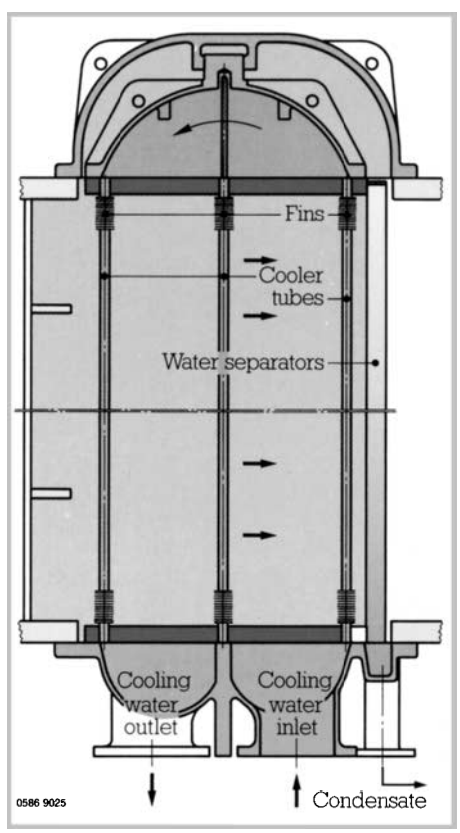


FIG. C-245 Two-pass version of the intercooler. Finned tubes and effective water separators contribute to the efficiency of the compressor. (Source: Sulzer-Burckhardt.)

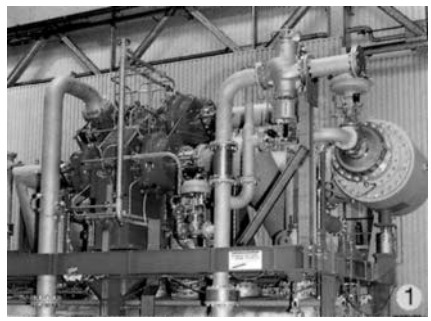


FIG. C-246 Six-stage oxygen integrally geared turbocompressor for a steel plant. (Source: Demag Delaval.)

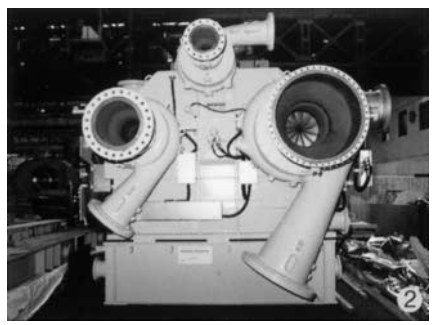


FIG. C-247 Dispatch preparation on a six-stage fuel gas compressor with dry gas seals for an LNG plant. (Source: Demag Delaval.)

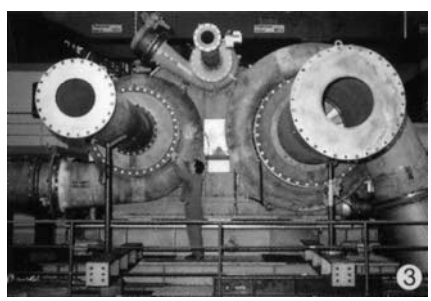


FIG. C-248 Test bed preparation of a five-stage integrally geared turbocompressor for an air separation unit. (Source: Demag Delaval.)

Compact integrally geared compressors. The CVK supplements the VK/HVK compressor range. Core units in four sizes and a wide range of pre-engineered options form the ideal basis to tailor a very cost-effective VK-gear–type compressor as a compact single-lift unit. (See Figs. C-264 and C-265.)

The application range. The CVK application range is subdivided into four sections, each of which is covered by one machine size. Depending on the required pressure ratio they can be designed with up to four stages.

Technical performance data (see Fig. C-266)

- Capacities from 10,000 to 32,000 m^2/h (6000 to 19,000 cfm)
- Suction pressure up to 1.1 bar (16 psi)
- Pressure ratio up to 20

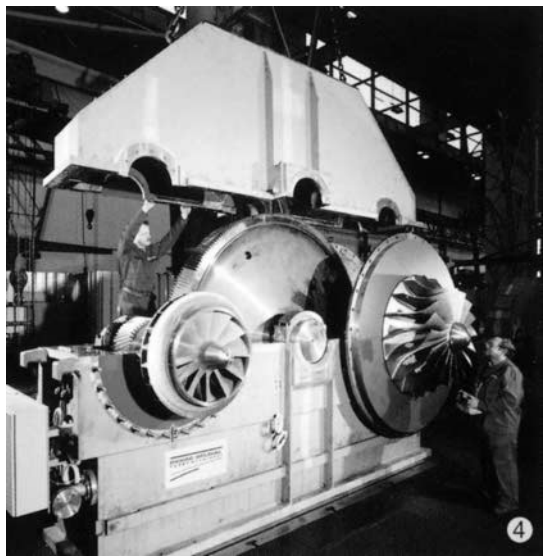


FIG. C-249 Assembly of an integrally geared turbocompressor for an integrated gasification combined cycle plant. (Source: Demag Delaval.)

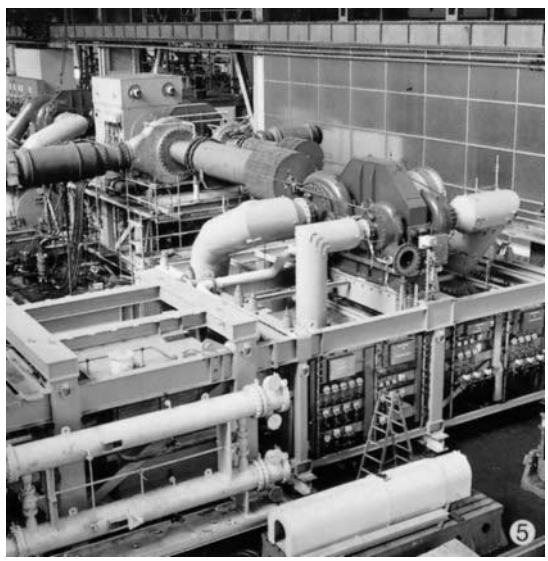


FIG. C-250 Packaging of two four-stage integrally geared turbocompressors with one expander for a PTA plant. (Source: Demag Delaval.)

Fields of application include

- Air separation
- Compressed air supply
- Hydrogen peroxide and more

Rotating Positive Displacement Compressors*

Rotary screw compressors and rotary piston blowers belong to the machinery group making up rotating positive-displacement compressors. Of these two machines,

* Source: Aerzener Maschinenfabrik, GmbH, Germany.

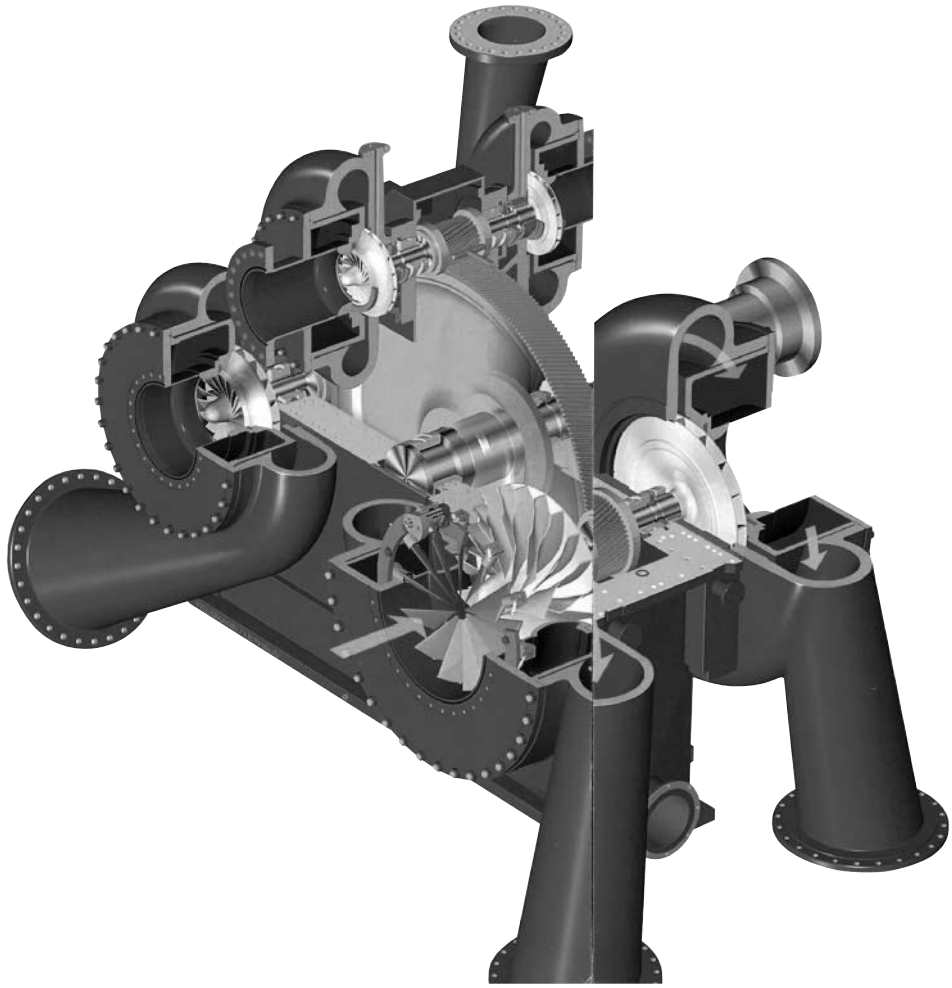


FIG. C-251 Internals of an integrally geared compressor. (Source: Demag Delaval.)

rotary screw compressors are primarily used in higher pressure air and process gas services. Rotary piston blowers are more typically used in lower pressure, high-volume applications. Both machines can be used as dry or wet fluid movers.

Rotating positive-displacement machines offer the same advantage as reciprocating positive-displacement equipment with regard to flow versus pressure relationships, i.e., nearly constant inlet flow volume under varying discharge pressure conditions. Also, positive-displacement machines do not have a surge limitation, which is to say, there is no minimum throughput requirement for these compressors.

The rotor tip speeds on rotary screw and rotary piston blowers are low; this allows for liquid injection and handling of contaminated gases. By design, the rotors are self-cleaning during operation, which is a significant advantage in dirty-gas services.

Rotary screw compressors are available in oil-free or oil-flooded construction. Fields of application for oil-free machines include all processes that cannot tolerate contamination of the compressed gas or where the lubricating oil would be contaminated by the gas. Oil-flooded machines can achieve slightly higher efficiencies and utilize the oil for cooling as well.

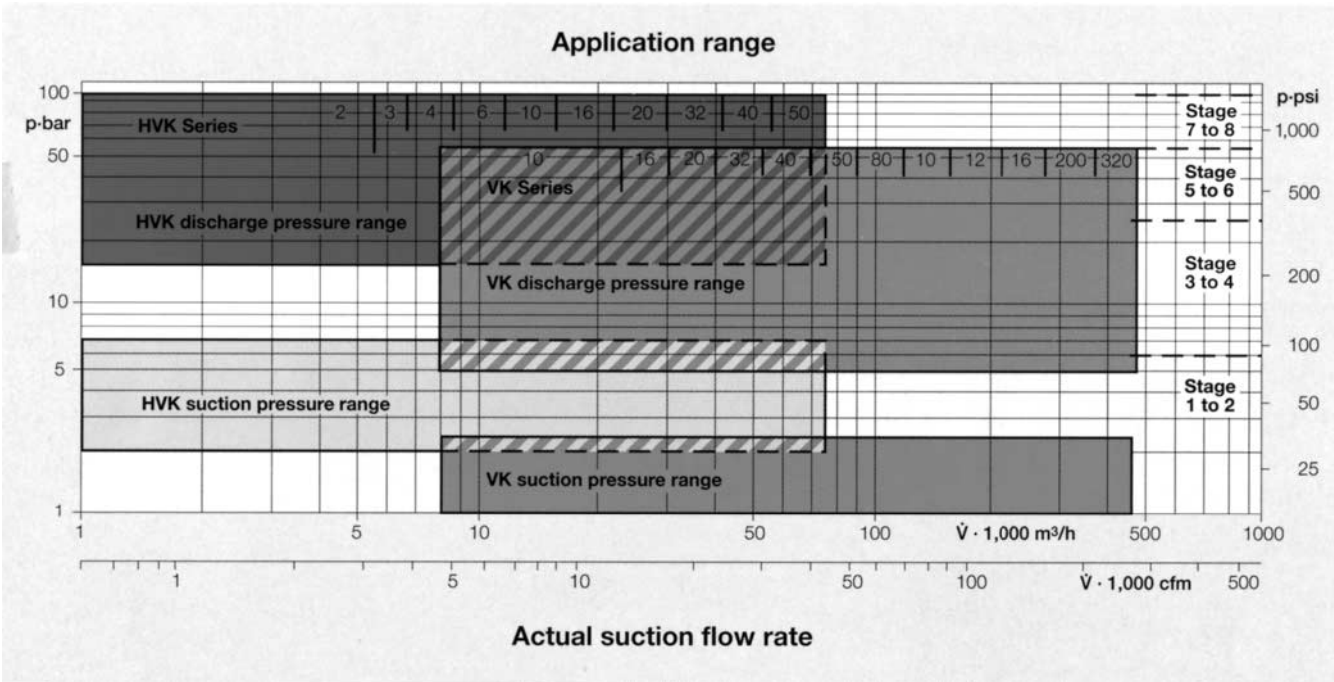


FIG. C-252 Operating envelopes for multistage integrally geared compressors. (Source: Demag Delaval.)

Properly designed rotary screw compressors are constructed with no metallic contact whatsoever inside the compression chambers, either between the rotors themselves or between these and the walls of the housing.

Although originally intended for air compression, rotary screw compressors are now compressing a large number of process gases in the petrochemical and related industries. These include air separation plants, industrial refrigeration plants, evaporation plants, mining, and metallurgical plants.

Most gases can be compressed: ammonia, argon, ethylene, acetylene, butadine, chlorine gas, hydrochloric gas, natural gas, flare gas, blast furnace gas, swamp gas, helium, lime-kiln gas, coking-plant gas, carbon monoxide gas, all hydrocarbon combinations, town gas, air/methane gas, propane, propylene, flue gas, crude gas, sulfur dioxide, oxide of nitrogen, nitrogen, styrene gas, vinyl chloride gas, and hydrogen gas can be found on the reference tabulations of experienced manufacturers.

Application limits for rotary screw compressors. Application limits for rotary screw compressors are given by the pressure and temperature ranges and by the maximum allowable speed of the machines.

Oil-free rotary screw compressors can be mechanically loaded with pressure differences up to 12 bar, and oil-flooded compressors up to 20 bar. Higher pressure differences are possible in special cases.

The maximum allowable compression ratio for one screw compressor stage that will not cause the final compression temperature to rise above the permitted value of 250°C will to a very large extent depend on the specific heat ratio c_p/c_v of the gas to be compressed. For example, where the specific heat ratio c_p/c_v equals 1.4, the maximum compression ratio would be approximately 4.5, and where the specific heat ratio c_p/c_v equals 1.2, the maximum compression ratio would be approximately 10 for one oil-free screw compressor stage.

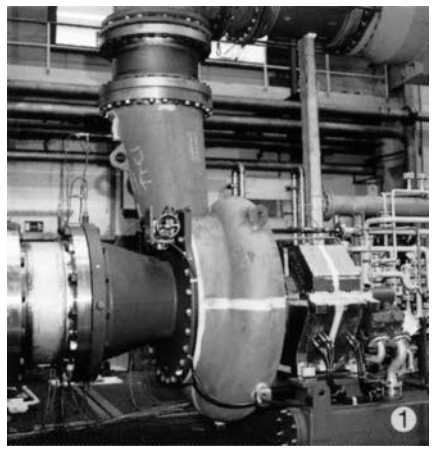


FIG. C-253 For a chemical plant during testing. (Source: Demag Delaval.)

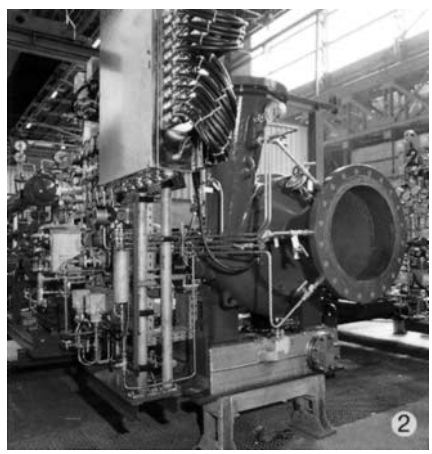


FIG. C-254 For nitrogen service in an ammonia plant. (Source: Demag Delaval.)

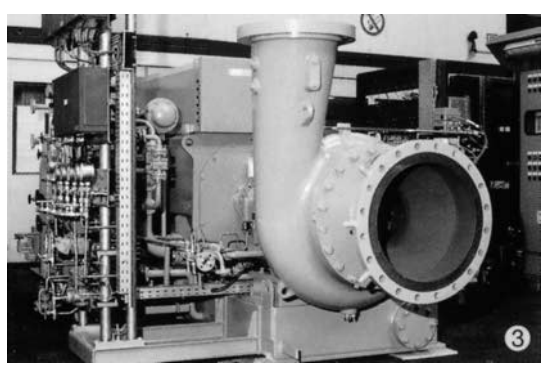


FIG. C-255 In preparation for shipment to a refinery. (Source: Demag Delaval.)

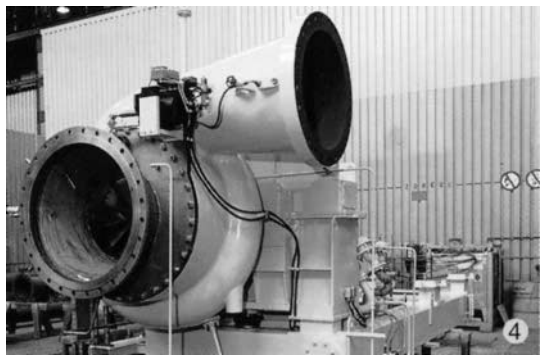


FIG. C-256 For a steel plant. (Source: Demag Delaval.)

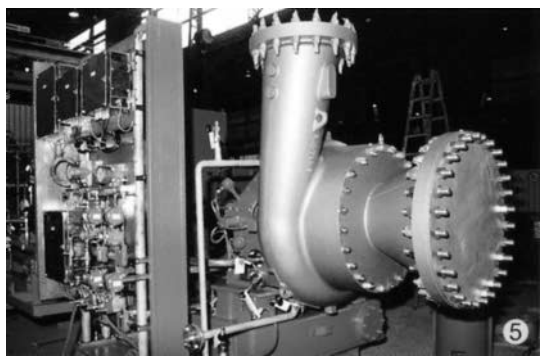


FIG. C-257 For oxygen service in an air separation unit. (Source: Demag Delaval.)

Multistage (multicasing) arrangements are not uncommon and can result in pressure ranges from approximately 0.1 bar absolute to 40 bar. Interstage cooling is used in many of these applications.

Depending on compressor size, speeds from 2000 to 20,000 rpm can be encountered. The limiting factor is typically the circumferential speed of the male rotor, which typically ranges from 40 to approximately 120 m/sec, and up to a maximum of 150 m/sec for very light gases.

Flow volumes up to 60,000 m^3/h can be accommodated in these compressors.

Reference and Additional Reading

1. Bloch, H., and Soares, C. M., *Process Plant Machinery*, 2d ed., Butterworth-Heinemann, 1998.

Rotary piston blowers*

Rotary piston, or lobe-type, blowers derive from the Roots compressor principle and have been built since 1864. They are used in a large variety of process plant applications, including pneumatic conveying of bulk materials, pressurized aeration of water at treatment plants, creation of vacuum, and gas movement in the petrochemical, pharmaceutical, and metallurgical industries. They range in size from fractional horsepower to literally hundreds of kilowatts.

*Source: Aerzener Maschinenfabrik, GmbH, Germany.

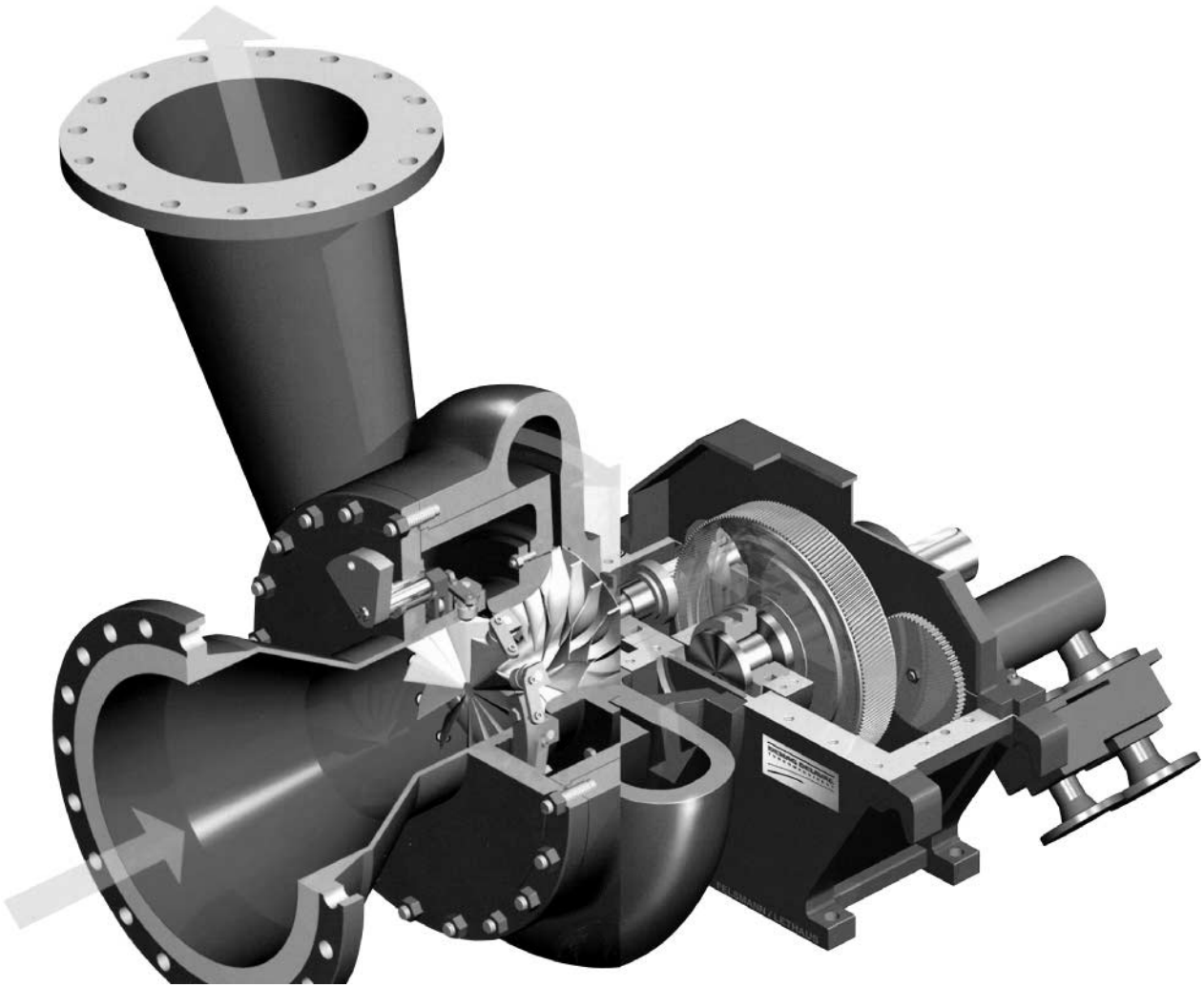


FIG. C-258 Internals of a single stage integrally geared compressor. (Source: Demag Delaval.)

Condensers*

The task of condensers is to condense the working fluid in a power plant cycle. The turbine exhaust steam condenses on the outside of the cooling tubes and the condensation heat is conducted to the cooling water inside the tubes. The condensate is collected and routed via the feedwater heaters and the feedwater line to the boiler where it again evaporates and is employed as operating steam to drive the turbine.

Via the turbine back pressure the performance of the condenser decisively influences the efficiency of the entire plant and with it the generator output (Fig. C-267).

In addition, the condenser represents a connecting link to the environment via the cooling water. This means that a condenser must fulfill not only the requirements of the power plant but also of environmental legislation.

*Source: Alstom. Adapted with permission.

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